

Issue Brief

AI on the Edge of Space

Securing Space Superiority and
Avoiding Surprise in Orbit

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Executive Summary

The U.S. Space Force faces growing threats from near-peer adversaries capable of targeting U.S. satellites, underscoring the need for enhanced space control capabilities. This paper examines how artificial intelligence can augment space domain awareness (SDA) and orbital warfare functions to help avoid operational surprise in orbit. Integrating AI, both on ground systems and onboard satellites, is essential to accelerating decision-making, enhancing satellite survivability, and maintaining domain knowledge in an increasingly contested environment.

This analysis reviews emerging AI applications upon two space mission areas and proposes additional areas for research. For the SDA mission area, it highlights the power of neural networks and explainable AI tools, such as Local Interpretable Model-Agnostic Explanations (LIME), to accelerate space object detection and improve sensor tasking efficiency. For the orbital warfare mission area, it explores how onboard AI agents can be applied to autonomously manage engagements through rendezvous and proximity operations (RPOs), optimize other satellite subsystems, and enable responsive payload tasking—within the constraints of satellite power and compute limitations. These findings are informed by recently published technical papers and defense policy documents.

The paper concludes with recommendations for responsible AI adoption into the above mission areas. These include the immediate adoption of some more mature SDA models, and procuring upgradeable satellite systems with sufficient onboard compute. This paper also recommends key policy considerations, such as defining boundaries for on-orbit autonomy, and establishing rigorous test and evaluation protocols to ensure transparent and auditable AI. In aggregate, implementing all or some of these efforts could significantly increase satellite survivability, and create opportunities to gain an algorithmically informed advantage to secure space superiority.

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Introduction

In recent years, space has become an increasingly contested, degraded, and operationally limited domain. Chief of Space Operations for the U.S. Space Force General B. Chance Saltzman has highlighted the dramatic rise in competitors' use of space and the exponential rise in hazards—such as orbital debris—that congest critical orbital regimes.¹ This evolving landscape underscores the importance of developing advanced technologies and associated concepts of operations to maintain a strategic advantage against adversary nations. Applications and fielding of such technologies to space control—activities required to contest and control the space domain—is inevitable.

Growing commercialization in the launch industry has fueled this domain's evolution.² In 2024, the number of active satellites in orbit surpassed 10,000, with more than 60 percent belonging to SpaceX's Starlink constellation.³ This explosive growth poses a serious challenge in accurately tracking, cataloging, and maintaining custody of satellites. Between 2020 and 2023, satellite conjunctions—or “near miss incidents”—rose by over 250 percent, requiring more frequent notifications for satellite owners to execute collision-avoidance maneuvers.⁴

Simultaneously, the miniaturization of processor technology has enabled satellites to carry more compute capabilities to orbit.⁵ While satellite maneuvering is not a new discipline, the development of more powerful onboard sensors, processors, and guidance, navigation, and control (GNC) systems are expanding the possibilities for new features, such as autonomous docking, evasion, or swarming capabilities.⁶ This poses a “dual-use” security risk, as systems capable of avoiding collisions can also be repurposed to maneuver satellites into close proximity with one another—potentially with destructive effects.

As satellites become more capable, major U.S. space stakeholders must integrate artificial intelligence (AI) to preserve operational advantage. In 2020, Russia demonstrated an ability to closely shadow and subsequently test space-based weapons in the vicinity of U.S. satellites.⁷ In 2022, China—using its Shijian-class satellites—demonstrated an ability to tow other satellites out of their mission orbits.⁸ These activities highlight an emerging threat: the ability of adversaries to exploit norms of behavior in space to gain first-mover advantage. Denying that advantage—through resilient, responsive systems—is central to deterrence in space.

AI can help U.S. space forces avoid operational surprise in orbit. While AI has already transformed other space functions—such as intelligence, surveillance, and

reconnaissance, two functions in particular stand to gain significant maturity: space domain awareness (SDA) and orbital warfare. These functions are critical to space control, and methodical application of AI to these areas would accelerate tactical decision-making, enhance satellite survivability, and increase orbital responsiveness. This paper examines how AI can augment SDA and orbital warfare capabilities, and outlines the policy recommendations needed to responsibly field such tools in future satellite architectures.

Accelerating the Sensemaking Process—AI for SDA

What Is SDA and How Is It Relevant to Space Control?

Space control consists of the activities required to contest and control the space domain.⁹ To develop such strategies, satellite operators must first understand the domain, including which orbits the engagement may happen in, what other satellites may be in the vicinity, and how naturally occurring or man-made hazards influence planning. To understand planning for space control, it helps to start from the end goal—locating a satellite in space—and working backward. Achieving comprehensive knowledge of an unknown satellite and its behavior requires several key pieces of information such as its position, capabilities, and intent.

SDA is the foundation of decision-making in space. It involves the effective identification, characterization, and understanding of any factor that could affect space operations and, by extension, the security of the U.S. space enterprise.¹⁰ SDA encompasses several subdisciplines, including detection, tracking, identification of resident space objects, characterization, threat warning and assessment, and data integration and exploitation.¹¹ Space operators use a combination of ground-based sensors (radar and optical sites) and space-based sensors (satellites with onboard signal or optical payloads) to obtain SDA on objects in space.

SDA underpins many other core space functions and is defined as an “enterprise” function for this reason.¹² For example, space electromagnetic warfare, satellite communications, missile warning, space-based sensing and targeting, and orbital warfare would not be possible without SDA. Today, the U.S. Space Force meticulously tracks and maintains custody of over 47,000 man-made objects in space and shares this crucial catalogue of data to other mission partners across the Department of Defense (DOD) and civil agencies such as the National Aeronautics and Space Administration (NASA).¹³ Furthermore, the United States—through the U.S. Space Force—is the only government entity that openly shares an unclassified space object catalog to users around the world.¹⁴

Military planners can leverage insights from space intelligence and SDA to build orbital warfare strategies, including concepts such as defending against or neutralizing adversary satellites. These plans typically involve the maneuvering of satellites to place them in a favorable position against a threat, enabling offensive posturing such as intelligence collection, or defensive posturing such as evasion of attacks.

Challenges with SDA

The accelerating pace of satellite launches and aging SDA architecture places growing strain on the U.S. Space Force Guardians performing this mission. Complete, constant global coverage is not always possible because of sensor gaps and tasking and scheduling conflicts.¹⁵ Operators must decide which objects in space to track, balancing priorities such as the age and quality of the object's last observed state, whether the object had maneuvered or not, and whether it is a high-priority target as directed by upper leadership.

Not all satellites maintain uninterrupted real-time telemetry with their respective ground station. The majority of satellites fly around the earth at a predictable velocity, requiring only spot checks using ground radars, optical sensors, or telemetry dishes to verify that the satellite is still maintaining its position and velocity. Because of the dynamic nature of orbits, a satellite may need to perform multiple passes over a ground sensor to establish an accurate track.

This opens the door to adversarial exploitation. Hostile actors could use sensor coverage gaps—for example, in regions of the southern hemisphere where the U.S. SDA architecture has fewer ground sensors—to attack U.S. satellites “in the dark.”¹⁶ Adversaries could also time the release of miniature payloads, such as debris or weapons, at opportune moments so that it would take days for operators to become aware of, establish a track on, and add to their catalogue. In 2023, General Saltzman lamented that it took up to seven days to forensically piece together a rocket body explosion that had fragmented one piece of space debris into four.¹⁷

Furthermore, with new “mega-constellations” such as Starlink on the rise, the challenge becomes even more difficult. According to the United Nations Office for Outer Space Affairs, the number of satellites launched in the last 36 months exceeds the total number launched in the first 60 years of space exploration.¹⁸ In 2023 alone, a record 2,895 objects were launched into space.¹⁹ This unprecedented volume places additional strain on the aging U.S. space surveillance and debris-tracking infrastructure, which must monitor an increasingly crowded and complex orbital environment.

In addition to the exponentially increasing number of objects launched into orbit, the complexity of the SDA mission is further compounded by the diversity of data sources (which originate from a wide variety of sensors) and the integration of mixed data from commercial systems, nongovernmental organizations, and partner nations. The steep learning curve associated with this mission also necessitates an acceleration of human-

machine teaming to meet operational demands. While AI tools cannot address all SDA challenges, process and hardware changes can be executed consistently to achieve tangible impact.

How AI Can Enhance SDA

Recent research converges on five SDA functions where AI can relieve bottlenecks: catalogue maintenance (including uncorrelated-track resolution); orbit determination; conjunction assessment; sensor tasking, scheduling, and prioritization; and data collection and integration. Across two studies, RAND researchers codified these categories after interviewing U.S. Space Force stakeholders. These functions are summarized below in Table 1.²⁰

Table 1: AI Support to SDA Functions

Function	AI Support to Function
Catalogue Maintenance and Uncorrelated Tracks	Rules-based expert systems (e.g., if-then-else rules), combined with neural networks, can accelerate the association of unknown objects to known orbit tracks. Expert systems capture analysts’ “if-then-else” rules for matching observations to catalog entries; neural networks handle the harder pattern-recognition cases.
Orbit Determination	Lightweight neural networks can reduce the time it takes to predict a satellite's future position in seconds instead of minutes to hours.
Conjunction Assessment	Two neural networks used in tandem can first narrow thousands of object pairs to the few that are most likely to collide, and then rapidly assign each pairing a risk score.
Sensor Tasking, Scheduling, and Prioritization	AI agents or hybrid models trained via reinforcement learning techniques can be optimized to reward successful detection of space objects, while balancing constrained sensor limitations. Trained models can predict upcoming collection taskings and schedule maintenance activities (such as sensor calibrations) in anticipation of the tasking.
Data Collection and Integration	Neural networks—particularly those that excel with missing or noisy data—can be used to harmonize disparate types of data from a variety of different sensors.

Maintaining the catalogue of space objects—and resolving the “**uncorrelated tracks**” of new objects launched into space—remains one of the most labor-intensive SDA tasks. While the U.S. Space Force expends considerable resources into maintaining an accurate catalogue, some objects in space may also move, break up, or for some other reason become disassociated with their “known track.” When these disassociations occur, it creates an event of interest, requiring operators to manually task sensors and re-correlate the object to its known track. This work is time sensitive, as uncorrelated objects carry a higher risk of collision.

Space operators may spend hours matching new observations to the correct catalogue entry whenever these events occur. One RAND study recommends codifying operator heuristics into a rules-based expert system to accelerate the process.²¹ Further, this expert system could be leveraged by passing only the most ambiguous cases to a neural network trained on years of historical correlations, to further aid operators in identifying uncorrelated tracks, or identify nefarious satellites prone to maneuvering. Unfortunately, re-correlation still hinges on how quickly the existing SDA architecture can perform orbit determination of the uncorrelated object.

Accurate **orbit determination** of an object’s location in space is therefore one of the fundamental building blocks for accurate SDA. When the network’s tracking data falls behind a satellite’s actual path, every follow-on task—warnings about possible close approaches, selecting sensors for observations, or sending updates to partner agencies—becomes less reliable. A second RAND study focuses on two lightweight neural networks that updated orbit states quickly enough to sustain in real-time operations.²² While one network forecasted an object’s future position, the other quantified how certain or uncertain that prediction was. By refreshing thousands of objects in near-real time and flagging only the handful that still needed a high-fidelity physics run, the system freed analysts to focus on difficult cases.

Those two networks, when used together, also streamlined **conjunction assessment**—predicting potential satellite collisions—to support collision avoidance. Because high-fidelity conjunction screening is a labor-intensive effort, it was impossible for space operators to screen every potential collision on a given day. Operators needed to filter down conjunctions that were highly likely. The uncertainty-aware network was able to rank those pairs by risk, producing a short, prioritized stack for operators. Only the highest-risk encounters triggered a deeper analytic run or a maneuver recommendation. By reducing false alarms, fewer “near miss” events are called. Satellites avoid unnecessary burns, extending their fuel and mission life.

Another cognitive bottleneck is **sensor tasking, scheduling, and prioritization**—deciding which radar, telescope, or antenna should point where, and when. One research team from the Massachusetts Institute of Technology trained a deep reinforcement-learning agent that earns rewards whenever its nightly schedule maximizes space object custody while respecting visibility windows and weather.²³ Another research team pioneered a hybrid model: it first uses machine learning methods to train a scheduling algorithm on small benchmark problems that have mathematically optimal answers, then subjects it to reinforcement learning in simulations until it was able to manage entire global sensor networks with diverse slew rates and calibration needs.²⁴ Both studies revealed different techniques to make significant gains in space object observations, resulting in more robust ground sensor coverage of the sky.

Lastly, the **data collection and integration** function presents a challenge because of the diversity of data formats originating from legacy (government-based) systems and new (often commercially based) SDA systems. Currently, there is no specifically trained model that can transform telemetry data from different sensors into a common format; however, this may be achievable with the appropriate training data set, if available.²⁵ Some types of neural networks, such as Bayesian or Hidden Markov models, excel with missing or noisy data.²⁶ Leveraging these models to infer missing fields or partial inputs could significantly improve SDA data streams. However, rigorous testing and human-in-the-loop oversight are essential to ensure these models neither “hallucinate” values nor hide critical outliers in downstream fusion systems. While these tools illustrate AI’s potential to unify disparate sensor streams, they also reveal a critical explainability gap that has yet to be addressed.

What Challenges Remain?

While much of the literature offers AI as a useful tool to enhance human-machine teaming and improve SDA, more work needs to be done to explain the logic behind models’ findings. Space operators may be less willing to adopt AI-informed outputs if they are unable to justify or explain how one sensor was selected to observe a space object over another. Embedding Local Interpretable Model-Agnostic Explanations (LIME) into these pipelines offers a solution. LIME works by lightly perturbing the input data, feeding it into the black box model, and observing how the outputs change.²⁷ These altered outputs are compared to the original results to surface which factors were most influential to the model’s decisions. For example, if a scheduling AI agent tasked an optical telescope to “observe a specific object at 12:00,” LIME might reveal that the decision was driven 50 percent by uncertainty in the space object’s orbit, 30

percent by the telescope's field of view, and 20 percent by an upcoming calibration window.

Such transparency delivers three concrete benefits. First, it satisfies DOD "Responsible AI" guidelines to ensure traceability, transparency, and auditability. Second, it reassures operators that they can override or veto an automated plan. And third, it builds confidence among military leaders and policymakers who must authorize AI adoption across the wider SDA portfolio.²⁸ By pairing these explanation layers with the performance gains documented in the above studies, U.S. Space Force operators could obtain faster, more efficient SDA while preserving the human insight and accountability that national-level decisions demand.

In addressing SDA challenges, this paper recommends expanding the use of AI to increase precision, timeliness, and efficiency. Table 1 identifies several areas wherein AI integration should be prioritized, with a focus on enhancing human-machine teaming, offloading cognitive burden, accelerating sensemaking, and supporting decision-making for small space operations crews. If the U.S. Space Force finds itself in a protracted conflict in space, consistently using these capabilities may become a necessity rather than a luxury.

Policymakers should consider the following actions:

- (1) Of the AI-enabled solutions identified, invest in the three more mature capabilities: (a) the layered expert-system and neural network for catalog maintenance and uncorrelated-track resolution, which has already reduced catalogue maintenance actions from hours to seconds; (b) the lightweight neural networks for rapid orbit updates and collision-risk triage, which outpace current operator workflows and can reduce false alarms; and (c) the deep reinforcement-learning scheduler (or its hybrid variant) for sensor tasking, which has demonstrably made sensor tasking more efficient under real-world constraints. Deploy each in an operational squadron or ground-station context to validate performance, integration requirements, and operator acceptance.
- (2) In concert with ongoing SDA upgrade efforts, ensure new systems are built with open interfaces, modular hardware and software components, and ample compute headroom to accommodate emerging AI tools. Building and embedding high-fidelity digital twins—complete with continuous-integration and continuous-deployment capabilities—to safely test, evaluate, validate, and verify algorithms via functional checkouts, adversarial-attack rehearsals, and human-machine teaming before any system-wide deployment is also essential.

- (3) Shift operator culture by launching targeted workshops, on-the-job training programs, and change-management initiatives to demystify new AI capabilities. This would ensure that the workforce is able and willing to adopt new AI tools. Introducing new AI tools may disrupt traditional and legacy workflows, and adoption may be further hampered by a lack of understanding, trust, or bandwidth to experiment with new systems.

In summary, this section outlines practical methods for applying AI to strengthen SDA, building on efforts identified in prior studies. As space becomes more contested, degraded, and operationally limited, SDA grows increasingly complex with the exponential rise in on-orbit objects. The widespread diffusion of AI across society underscores an imperative to extend the technology into space. Understanding the domain and the satellites that maneuver within it is the first step; applying that understanding to better maneuver and manipulate satellites is the next.

Gaining Maneuver Advantage—AI for Orbital Warfare

What Is RPO and Why Is It Relevant to Space Control?

The act of maneuvering a satellite against other satellites is known as rendezvous and proximity operations (RPO). Rendezvous refers to aligning two separate satellites into the same orbital plane, shape, and size. Proximity operations involve bringing them together to perform actions such as station-keeping, formation-flying, inspection, or docking. As enabling technologies mature, RPOs have become faster, more precise, and safer. Today, satellites use a combination of onboard sensors—such as cameras and LiDAR—and external data relayed via the aforementioned SDA network to determine their relative position to other spacecraft.

RPOs have been performed for decades. The first successful RPO occurred in 1965 between Gemini 6A and Gemini 7: Gemini 7 was launched and placed in orbit as a target and Gemini 6A was launched 11 days later, ultimately approaching as close as one foot away as the two vehicles orbited Earth.²⁹ Since then, RPO capabilities have evolved significantly. Of the more than 80 space-faring nations today, 12 are actively developing counterspace capabilities—including tactics, techniques, and procedures involving RPO maneuvers—with several having demonstrated them in orbit.³⁰

Satellites have long been considered extensions of national power, offering communication, intelligence, navigation, and other military capabilities for terrestrial operations on land, in the air, or at sea. RPOs conducted for military purposes—whether for intelligence collection, offensive action, or defense—are generally executed away from the public eye, without fanfare or alarm. Announcing such maneuvers in advance may risk alerting adversaries, giving them time to reposition or conceal sensitive payloads onboard.³¹

This norm is rapidly changing, and there has been much more focus on such maneuvers. In the near future, satellites may conduct “dogfights” in space—the capability to do so already exists, though it is not always characterized in this way.³² One such example is China’s Shijian class of experimental satellites, which demonstrated advanced RPO maneuvers in 2022 to grapple a defunct satellite, towing it to a graveyard orbit.³³ Though touted as “space debris mitigation,” such dual-use capabilities could easily be repurposed to disable or degrade operational satellites in a wartime setting. The same technologies that enable safe RPO can also enable orbital warfare.

Historically, RPOs have been used for civil or commercial purposes—such as docking for human spaceflight, on-orbit refueling, and debris removal. Today, RPO-capable satellites also serve as space-based sensors in the SDA architecture, maneuvering to track, follow, and inspect high-interest objects in space. However, in a conflict, RPO activity is likely to increase, particularly in low earth orbit (LEO), or strategic regions of geosynchronous orbit (GEO) where national security satellites often reside.

This potential escalation carries implications for military and civilian users alike. Communications, remote sensing, and precision timing and navigation services—in some cases delivered via dual-use satellites—could be disrupted. Stakeholders must understand these risks and explore how AI-enabled autonomy could help mitigate disruptions in contested environments.

Challenges with RPO

Though RPOs have been practiced for decades, their operation remains highly complex and typically requires skilled human operators and subject matter experts to execute. Operators must first set favorable conditions for a successful RPO by assessing the status of both spacecraft—the “attacker” and the “target”—and determining whether the target is cooperative or noncooperative. Each category of targets requires different planning methodologies and engagement strategies.

Cooperative targets are typically willing receivers of an RPO. In 2020, Intelsat 901 became the first satellite in GEO to receive a cooperative RPO, partnering with Northrop Grumman’s Mission Extension Vehicle-1 (MEV-1).³⁴ MEV-1 docked with Intelsat 901 to extend the 18-year-old satellite’s lifespan by five years. The success of the mission was enabled by close collaboration between Northrop Grumman and Intelsat, whose teams shared satellite and telemetry data in real time to ensure the mission was a success. In 2025, MEV-1 successfully moved Intelsat 901 into a graveyard orbit, replicating China’s Shijian 21 mission three years earlier.³⁵

Noncooperative targets, by contrast, include nonfunctional satellites (i.e., space debris or “gray” satellites with limited or no maneuverability) as well as adversarial satellites exhibiting evasive or potentially offensive behavior. These satellites are incentivized to avoid participation in an RPO engagement to protect sensitive onboard capabilities. In 2019, such an engagement occurred between USA-245—an intelligence satellite owned by the National Reconnaissance Office—and Cosmos 2542, a Russian satellite widely speculated to be an anti-satellite platform. Cosmos 2542 was launched into the same orbit as USA-245 and pursued the satellite for days, coming as close as 20 kilometers of the U.S. satellite.³⁶ In 2020, United States Space Command observed

Cosmos 2542 releasing a sub-satellite—labeled Cosmos 2543—which also trailed USA-245 in close formation.³⁷ The aggressive behavior of Cosmos 2542 and 2543 prompts a wider discussion on the need for satellite evasion or defensive systems.

Operators must also consider platform constraints when selecting sensors to support RPO missions. Different sensor configurations may consume different amounts of power and may only be suitable for certain engagement ranges. For example, MEV-1 used a LiDAR system, which provided highly accurate measurements at close range but consumed more power than its optical camera suite.³⁸ However, LiDAR is ineffective until the spacecraft is in close proximity to the target. Operators would have to balance the use of both systems, using the camera suite to identify and track the target at longer distances before switching to LiDAR for final approach. Managing such trade-offs—particularly under power and compute constraints—is a prime example of when human-machine teaming can optimize performance.

Trajectory planning also presents risk. If a noncooperative target attempts to evade, it could inadvertently create a collision scenario. Such events may unfold much faster than operators are able to detect, interpret, and respond—especially if adversaries maneuver within the SDA “dark zones” described earlier in this paper. To support satellite operators in these high-stakes, fast-paced environments, this paper identifies several areas in which AI can enhance satellite function—improving sensing, autonomous decision-making, or responsiveness during complex RPO missions.

How AI Can Enhance Orbital Warfare

Historically, most legacy satellites relied on simple, rules-based software because of limited onboard computing power. The capability to host more complex algorithms was limited by the hardware requirements needed to run them. However, with advances in hardware miniaturization and increasingly efficient algorithms, the feasibility of running AI onboard satellites have never been greater. Table 2 lists various functions that onboard AI can support.³⁹

Table 2: AI Support to Orbital Warfare

Function	AI Support to Function
Object and Movement Detection	Onboard computer vision-based neural networks can process images in real time to detect, classify, and track objects or targets of interest. Autonomous processing reduces the need for constant data-downlink, enabling faster decision-making to cue subsequent processes or tasks.
Autonomous Guidance, Navigation, and Control (GNC)	AI-driven predictive algorithms can take real-time data to autonomously develop and execute satellite maneuvering, orbit maintenance, and collision avoidance. Additionally, autonomous models can optimize operations by planning fuel-efficient RPO trajectories, reducing the cognitive burden on human operators.
Data Filtering and Prioritization	Predictive algorithms combined with data filtering methods can prioritize critical data for transmission, to improve operational efficiency, and to reduce latency in contested or electromagnetically degraded environments. AI models can autonomously identify and filter out only the most critical data to be downlinked, reducing bandwidth usage.
Autonomous Bus Subsystem Management	AI agents can manage satellite bus subsystems by optimizing power, thermal, and attitude control; detecting and responding to faults or anomalies; and reallocating resources in real time. These efficiencies extend mission life, reduce operator burden, and enhance resilience in dynamic environments.
Dynamic Communications Management	Neural networks and AI agents can quickly resolve link failures and ensure reliable data transmission without human intervention by dynamically allocating bandwidth or switching frequencies to avoid interference, signal degradation, or jamming.
Autonomous Payload Scheduling and Prioritization	Onboard AI agents can schedule and prioritize payload operations (e.g., imaging, LiDAR, docking mechanisms) based on available telemetry, power levels, and mission objectives to maximize task efficiency and mission output, reducing reliance on manual taskings when out of view from ground terminals.

Space-based systems operate under harsh conditions and tight power constraints, requiring trade-offs between computing capability, energy consumption, and radiation tolerance. Unlike the powerful chips used in terrestrial data centers, space-grade processors are less capable but must be hardened against cosmic rays, solar particles, and ionizing radiation. While larger AI models may deliver better performance, they often consume too much power. As a result, successful onboard compute depends on two key factors: selecting radiation-tolerant AI processors and designing models that operate efficiently within strict power and thermal limits. This section examines models first.

Computer vision—a subfield of AI focused on training models to interpret imagery—has become central to space-based **object and movement detection**. Accurate detection underpins target custody, helps derive target intent and activity, and supports **GNC** against the target. A variety of computer vision techniques—most notably convolutional neural networks—contribute to these functions, but the tasks become markedly harder when the satellite must track noncooperative or maneuvering targets. One promising area of research is in spacecraft “pose estimation” as a potential augmentation to existing computer vision research.⁴⁰

Originally developed for tracking human motion, pose estimation is now being adapted toward RPO use.⁴¹ In particular, one team at Stanford University developed an AI-based vision system trained entirely on synthetic images, simulating spacecraft orientations at various angles. The model weights were then rigorously fine-tuned under realistic lighting and sensor noise conditions in a lab. Results showed that the model could reliably track tumbling or maneuvering satellites it has never seen before, with minimal ground operator intervention—marking a significant step forward in RPO autonomy.⁴²

Onboard AI may also help **autonomously filter and prioritize data** for transmission to ground stations—a critical function in bandwidth-limited or electromagnetically contested environments. One recent study from the Advanced Maui Optical and Space Surveillance Technologies Conference demonstrated this by deploying a lightweight neural network on a space-based processor to detect objects in captured imagery. If objects were detected, the results would be downlinked. However, rather than downlinking full-resolution images, the system transmitted only bounding boxes and confidence scores. This reduced bandwidth usage by 93 percent, enabling faster data delivery and lowering communication overhead—key advantages in time-sensitive or electromagnetically contested situations.⁴³ This capability—if adopted—could accelerate decision-making during RPO engagements, enabling inspector satellites to more quickly investigate features on a target satellite, or evade oncoming debris.

While many studies have focused on the use of AI for improved object detection and GNC (particularly for RPOs), comparatively little research has been directed into AI-enabled **management of satellite bus subsystems, dynamic communication management, or autonomous payload scheduling**. A satellite consists of two primary components: the payload (such as imagers, transmitters, robotic arms, or other mission-critical equipment), and the bus (which includes power, thermal control, communications, propulsion, and attitude control subsystems essential for supporting the payload). The bus is just as vital as the payload in autonomous satellite defense, and the application of AI to its functions presents significant possibilities for increased efficiency and survivability.

In scenarios in which a satellite faces a direct attack or disruptive event, human operators may lose the telemetry link, leaving the satellite isolated and vulnerable. In such cases, onboard AI agents could autonomously detect anomalous activity—such as losing custody of a tracked target satellite or encountering jamming of communications signals—and dynamically reallocate power away from nonessential subsystems toward critical defensive or evasive maneuvers. For example—if a satellite’s proximity operations were interrupted, the satellite could autonomously attempt to re-acquire its target, while simultaneously managing its communications systems to reestablish contact with ground-based operators. These layered, autonomous functions reduce dependence on ground control, ensuring satellite survivability in environments where timely human intervention might not be possible.

What Challenges Remain?

Given these capabilities, continued research and development in onboard AI is essential. This paper recommends focused research and investments to enhance satellite resilience and ensure mission continuity in contested or electromagnetically degraded environments. Priority functions include real-time fault mitigation (e.g., rerouting power or isolating damaged subsystems), dynamic bus subsystem management, and autonomous reprioritization of payload tasks in response to evolving threats or mission demands. These capabilities would reduce reliance on continuous ground control and allow satellites to maintain operations under adverse conditions. If the U.S. Space Force intends to field resilient, responsive constellations for future conflicts, investment in these technologies is imperative.

Despite advances in onboard AI, a critical policy gap remains: there are no clear definitions for the operational boundaries of autonomous satellite behavior. Current guidance lacks detailed specifications for computational benchmarks, performance metrics, and criteria for test, evaluation, validation, and verification (TEVV) of space-

based AI. At a broader level, effective governance must clearly establish how much autonomy the DOD is willing to delegate to satellites. If adopted, should AI agent autonomy be limited to self-defense, or extended to GNC throughout a satellite's lifespan? These questions remain unresolved—and carry significant operational implications.

Policymakers should consider the following actions:

- (1) Clearly define the acceptable boundaries of on-orbit autonomy, specifying conditions under which satellites may operate without direct human intervention. This includes clarifying whether autonomy is strictly defensive or extends to broader operational contexts such as routine GNC, object tracking, and payload management.
- (2) Publish detailed technical performance standards, specifying computational benchmarks, allowable power consumption, radiation tolerance, and environmental resilience necessary for effective space-based AI systems. Clear technical standards will facilitate robust system design and reliable performance in harsh space environments.
- (3) Implement comprehensive, space-specific TEVV processes to ensure AI systems are consistently reliable and trustworthy. Rigorous verification standards tailored to onboard AI systems will help identify vulnerabilities before deployment, increasing confidence in the performance of autonomous systems.
- (4) Formalize explicit AI acquisition guidelines tailored specifically for satellite systems. These guidelines must detail operational requirements, risk management strategies, and integration considerations, equipping acquisition professionals to responsibly procure and field AI-enabled satellite capabilities.

As the U.S. Space Force executes its AI Strategic Action Plan, downstream policies must explicitly incorporate these recommendations to form a robust and comprehensive governance framework for responsible AI deployment.⁴⁴ Existing guidance, such as the DOD's Directive 3000.09 for autonomy in weapon systems, and the Responsible AI Strategy and Implementation Pathway, provide a starting point but currently lack the granularity that space acquisition professionals need.⁴⁵ The satellite functions outlined in this section offer a practical foundation upon which detailed AI procurement and operational guidelines can be derived. Without implementing such guidance, the United States risks either underutilizing a critical capability or deploying it in ways that introduce unseen vulnerabilities into its satellite systems.

Onboard Compute Is Limited by Hardware...

...but This Is Quickly Changing

Advances in AI model efficiency, hardware miniaturization, and structured training datasets have ushered in a new era of onboard autonomy. While significant efforts have been applied toward developing more capable onboard models, attention must also be focused on hardware. Not all processors are suitable for space. Radiation-hardened chips are required, and these typically feature greater redundancy but significantly less computational power than their Earth-bound counterparts.

BAE Systems' RAD 5545 chip boasts more than twice the clock speed, eight times the memory, and 1,000 times the floating-point performance of its predecessor, the processor used in the Mars Curiosity Rover.⁴⁶ These enhancements are critical for supporting onboard AI. Meanwhile, commercial off-the-shelf alternatives such as AMD's Versal XQR and NVIDIA's Jetson Orin NX and AGX Orin offer comparable AI capabilities at a fraction of the RAD 5545's cost.⁴⁷ As radiation-hardened (or harden-ready) chips become more widely available, the barriers to deploying AI-driven autonomy in space are rapidly falling.

In orbital warfare, onboard AI is especially valuable in latency-critical scenarios in which milliseconds can determine the life or death of a satellite. Autonomous systems can employ defensive measures or accomplish mission objectives without receiving commands from the ground. If a satellite needs to evade an oncoming threat—an anti-satellite missile, for example—having onboard compute could enable the satellite to maneuver or defend itself even if telemetry with a ground station were disrupted. As space becomes more contested, survivability in electromagnetically degraded environments will be a defining requirement, especially during wartime.

What Is Possible?

Onboard AI opens the door to novel tactics, techniques, and procedures for satellite operations. Much like the F-35's experimentation with collaborative aircraft, AI-enabled satellites could fly in close formation—enabling concepts such as defender satellites to protect high-value assets in orbit.⁴⁸ In contested environments, these formations could detect, assess, and respond to nearby threats—such as repositioning a defender satellite between an approaching object and a protected platform—without

waiting for ground commands. Such techniques could increase the survivability of older on-orbit satellites that do not possess similar capabilities.

Looking ahead, if onboard AI models can be securely updated throughout a satellite's operational life, a wide range of dynamic use cases becomes possible. In response to new threats launched into orbit, the U.S. Space Force may not need to deploy new hardware. Instead, ground teams could retrain AI models on updated data and uplink them to satellites already in orbit—reprogramming their behavior in flight. This approach could dramatically extend satellite utility, reduce or eliminate costs, and accelerate adaptation to newly-discovered threat conditions.

Over time, such capabilities could support a shift toward more agile constellations in which spacecraft autonomously re-task themselves based on adversary behavior, mission demands, or emergent threats. Satellites able to autonomously operate in contested environments will demand appropriate hardware. Acquisition professionals should account for these future-facing needs when drafting program requirements or evaluating vendors, ensuring there is adequate compute capacity to host increasingly complex models onboard satellites.

Conclusion

As the space domain grows more contested, degraded, and operationally constrained, the need to integrate artificial intelligence into both SDA and orbital warfare functions has never been more urgent. While embedding AI systems—both onboard satellites and within ground-based SDA infrastructure—may introduce new attack surfaces, such as data poisoning or evasion attacks vulnerabilities, these risks can be mitigated.

⁴⁹ With clear guardrails—performance standards, explainability requirements, rigorous TEVV, and hardened data pipelines—the U.S. Space Force can harness AI to accelerate tactical decision-making, enhance satellite survivability, and reduce the risk of operational surprise in orbit.

For SDA, this paper recommends a three-pronged approach. First, deploy mature AI tools with proven operational performance—such as lightweight neural networks for real-time orbit determination and conjunction triage. Second, procure high-fidelity simulation and digital-twin testbeds, equipped with continuous-integration and continuous-deployment pipelines, to validate algorithms across varied orbital scenarios. These virtual environments facilitate functional checkouts of new algorithms, adversarial-attack rehearsals, and human-machine teaming exercises without endangering live assets. Third, implement targeted training programs that build operator trust in AI tools. These programs should incorporate explainable-AI techniques (e.g., LIME) to clarify why models flag certain conjunction risks or recommend specific taskings—preserving human oversight and accountability.

For orbital warfare functions, AI-enabled autonomy offers a forward leap in capabilities. Research and development should prioritize pose-estimation algorithms for refining navigation during high-risk RPOs; and AI agents for autonomous bus subsystem management; dynamic communications capabilities to circumvent jamming or degraded bandwidth; and onboard schedulers that reprioritize payload tasks in real time. To ensure these systems perform reliably in contested environments, policy must also define clear autonomy boundaries, establish technical benchmarks for space-based AI, and implement tailored TEVV and acquisition processes for both hardware and software.

Further, several cross-functional enablers should be considered:

(1) **Ethical, Legal, and Norms Frameworks**

The growing potential of AI agents for satellite operations elicits profound treaty and arms-control considerations. The United States should lead the world in

codifying norms of behavior for the use of AI in space, and should collaborate on new multilateral standards for AI-enabled space operations.

(2) Joint and Combined Interoperability

Space is inherently multinational. In developing its SDA architecture, the DOD should ensure data compatibility and system interoperability within the Joint force, and amongst partner nations. The U.S. Space Force and combatant commands should also conduct joint exercises to validate AI-informed concepts of operation under coalition SDA architectures, validating the operational utility of these systems.

(3) Metrics and Cost-Benefit Analysis

Clear, quantitative metrics—such as “mean time to replan RPO trajectory,” “mean time to correlate new debris,” and “mean time to perform conjunction analysis”—will inform acquisition requirements and drive U.S. Space Force budget decisions. Regularly published scorecards will demonstrate the value of AI investments and guide congressional resource allocation.

(4) Model Sustainment and Software Updates

Satellites remain in orbit for years, so their embedded AI models must evolve. Satellite command and control architectures must accommodate secure, authenticated, ground-to-orbit software and model updates. In doing so, satellites can adapt to emerging threats and benefit from algorithmic improvements, prolonging their usefulness in orbit.

(5) Workforce Development and Talent Management

Cutting-edge autonomy demands operators who understand and trust AI. In addition to previously mentioned exercises and training to facilitate the adoption of AI-enabled systems, the DOD should continue to collaborate within the U.S. AI ecosystem to take advantage of technological breakthroughs emerging from academia and industry. Additionally, the U.S. Space Force should consider establishing specialized career tracks to mature a cadre of AI experts.

Each of these enablers may require additional research and will benefit from future studies. By weaving together technical rigor, policy foresight, and organizational readiness, the U.S. Space Force can move beyond proof-of-concepts to operationally mature autonomous systems. These capabilities will not only enhance SDA and bolster orbital warfare—they will also prove to be decisive enablers to secure U.S. space superiority.

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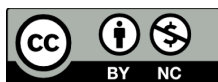
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Endnotes

¹ Joseph Clark, "Space Force General Outlines U.S. Approach to Maintaining Space Superiority," U.S. Department of Defense, March 28, 2024, <https://www.defense.gov/News/News-Stories/Article/Article/3723145/space-force-general-outlines-us-approach-to-maintaining-space-superiority/>.

² Corey Crowell and Sam Bresnick, "Defending the Ultimate High Ground: China's Progress Toward Space Resilience and Responsive Launch," Center for Security and Emerging Technology, July 12, 2023, <https://cset.georgetown.edu/publication/defending-the-ultimate-high-ground/>.

³ Eric Mack, "There Are 10,000 Active Satellites in Orbit. Most Belong to Elon Musk," *Forbes*, July 19, 2024, <https://www.forbes.com/sites/ericmack/2024/07/19/theres-now-10000-active-satellites-in-orbit-most-belong-to-elon-musk/>.

⁴ Statistics were obtained through an interview with Dr. Alinda Mashiku, NASA's Conjunction Assessment Risk Analysis Program Manager on May 19, 2025. National Aeronautics and Space Administration, "Unique Conjunction Events," NASA, February 2, 2022, <https://www.nasa.gov/cara/unique-conjunction-events/>.

⁵ Kevin Walker, "Miniaturization—Driving the next Generation of Space Technology," *Benchmark*, June 11, 2024, <https://www.bench.com/setting-the-benchmark/miniaturization-is-accelerating-modern-aerospace-technology>.

⁶ Tara Friesen, "Swarming for Success: Starling Completes Primary Mission," NASA, May 29, 2024, <https://www.nasa.gov/general/swarming-for-success-starling-completes-primary-mission/>.

⁷ U.S. Space Command Public Affairs Office, "Russia Conducts Space-Based Anti-Satellite Weapons Test," U.S. Space Command, July 23, 2020, <https://www.spacecom.mil/Newsroom/News/Article-Display/Article/2285098/russia-conducts-space-based-anti-satellite-weapons-test/>.

⁸ Brett Tingley, "A Chinese Satellite Just Grappled Another and Pulled It out of Orbit," *The War Zone*, January 27, 2022, <https://www.twz.com/44054/a-chinese-satellite-just-grappled-another-and-pulled-it-out-of-orbit>.

⁹ Space Training and Readiness Command (STARCOM), "Space Force Doctrine Document – 1 (SFDD-1)," Space Training and Readiness Command, April 3, 2025, [https://www.starcom.spaceforce.mil/Portals/2/Space Force Doctrine Document 1 FINAL_4Apr25.pdf](https://www.starcom.spaceforce.mil/Portals/2/Space%20Force%20Doctrine%20Document%201%20FINAL_4Apr25.pdf).

¹⁰ Space Operations Command, "Mission Delta 2 - Space Domain Awareness," Space Operations Command (SpOC), April 2025, <https://www.spoc.spaceforce.mil/about-us/fact-sheets/display/Article/3878122/mission-delta-2-space-domain-awareness>.

¹¹ Li Ang Zhang, Krista Langeland, Jonathan Tran et al., "Artificial Intelligence and Machine Learning for Space Domain Awareness: Characterizing the Impact on Mission Effectiveness" (RAND Corporation, November 21, 2024), https://www.rand.org/pubs/research_reports/RRA2318-1.html.

¹² Space Training and Readiness Command (STARCOM), “Space Force Doctrine Document – 1 (SFDD-1).”

¹³ Space Operations Command, “18th Space Defense Squadron,” Peterson and Schriever Space Force Base, January 2023, <https://www.petersonschriever.spaceforce.mil/About-Us/Fact-Sheets/Display/Article/3016228/18th-space-defense-squadron/>.

¹⁴ Dr. Mariel Borowitz, “Space Situational Awareness: Guiding the Transition to a Civil Capability,” Written Testimony to the Senate Committee on Science, Space, and Technology, 117th Congress, April 29, 2022, <https://www.congress.gov/117/meeting/house/114695/witnesses/HHRG-117-SY16-Wstate-BorowitzM-20220512.pdf>; Mikayla Easley, “Space Force Launches New Effort to Share Unclassified Threat Data with Commercial Industry,” *DefenseScoop*, April 9, 2025, <https://defensescoop.com/2025/04/09/space-force-share-unclassified-threat-data-commercial-industry-orbital-watch/>.

¹⁵ Theresa Hitchens, “SPACECOM Needs More Data, Sensors to Track On-Orbit Threats,” *Breaking Defense*, September 17, 2021, <https://breakingdefense.com/2021/09/spacecom-needs-more-data-sensors-to-track-on-orbit-threats/>.

¹⁶ Theresa Hitchens, “Space Force Looking to Fill Tech ‘Gaps,’ and ‘Evolve’ Space Domain Awareness: Saltzman,” *Breaking Defense*, September 20, 2023, <https://breakingdefense.com/2023/09/space-force-looking-to-fill-tech-gaps-and-evolve-space-domain-awareness-saltzman/>.

¹⁷ Josh Luckenbaugh, “Space Force Still Dodging Chinese Satellite Debris,” *National Defense Magazine*, December 15, 2023, <https://www.nationaldefensemagazine.org/articles/2023/12/15/space-force-still-dodging--chinese-satellite-debris>.

¹⁸ United Nations Office for Outer Space Affairs, “Cumulative Number of Objects Launched into Space,” *Our World in Data*, 2025, <https://ourworldindata.org/grapher/cumulative-number-of-objects-launched-into-outer-space?time=earliest..2017>; A cumulative total of 7,997 satellites were launched into space between 1957 and 2017. This number is compared to 8,221 satellites launched into space between the beginning 2022 and end of 2024.

¹⁹ Global Change Data Lab, “Annual Number of Objects Launched into Space,” *Our World in Data*, 2025, <https://ourworldindata.org/grapher/yearly-number-of-objects-launched-into-outer-space?time=latest>.

²⁰ List of tasks obtained from a 2024 RAND study focused on AI/ML applications to SDA. Zhang et al., “Artificial Intelligence and Machine Learning for Space Domain Awareness.”

²¹ Zhang et al., “Artificial Intelligence and Machine Learning for Space Domain Awareness.”

²² Jonathan Tran, Prateek Puri, Jordan Logue et al., “Artificial Intelligence and Machine Learning for Space Domain Awareness: The Development of Two Artificial Intelligence Case Studies” (RAND Corporation, September 30, 2024), https://rand.org/pubs/research_reports/RRA2318-2.html.

- ²³ Peng Mun Siew and Richard Linares, "Optimal Tasking of Ground-Based Sensors for Space Situational Awareness Using Deep Reinforcement Learning," *Sensors* 22, no. 20 (October 16, 2022): 7847, <https://doi.org/10.3390/s22207847>.
- ²⁴ Neil Dhingra, Cameron Dejac, and Clayton Mcguire Auria, "Machine Learning for Space Domain Awareness Sensor Scheduling," *Advanced Maui Optical and Space Surveillance Technologies Conference*, 2024, <https://amostech.com/TechnicalPapers/2024/Poster/Dhingra.pdf>.
- ²⁵ Emelia Probasco, Helen Toner, Matthew Burtell, and Tim Rudner, "AI for Military Decision-Making: Harnessing the Advantages and Avoiding the Risks," Center for Security and Emerging Technology, April 2025, <https://cset.georgetown.edu/publication/ai-for-military-decision-making/>.
- ²⁶ Coursera Staff, "A Beginner's Guide to the Bayesian Neural Network," Coursera, September 18, 2024, <https://www.coursera.org/articles/bayesian-neural-network>.
- ²⁷ Marco Tulio Ribiero, Sameer Singh, and Carlos Guestrin, "'Why Should I Trust You?': Explaining the Predictions of Any Classifier," arXiv preprint arXiv.1602.04938 (2016), <https://arxiv.org/abs/1602.04938>.
- ²⁸ Chief Digital and Artificial Intelligence Office, "Responsible AI," Department of Defense, 2020, <https://www.ai.mil/Initiatives/Responsible-AI/>.
- ²⁹ David Williams, "Gemini 6A," NASA Space Science Data Coordinated Archive, October 28, 2022, <https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1965-104A>; David Williams, "Gemini 7," NASA Space Science Data Coordinated Archive, October 28, 2022, <https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1965-100A>; John Uri, "55 Years Ago: The Spirit of 76 - the First Rendezvous in Space," NASA, December 8, 2020, <https://www.nasa.gov/history/55-years-ago-the-spirit-of-76-the-first-rendezvous-in-space/>.
- ³⁰ Victoria Samson, "Breaking the Impasse over Security in Space" (Arms Control Association, September 2022), <https://www.armscontrol.org/act/2022-09/features/breaking-impasse-over-security-space - endnote01>; Victoria Samson and Laetitia Cesari, "Global Counterspace Capabilities: An Open Source Assessment," Secure World Foundation, April 2025, https://swfound.org/media/208099/swf_global_counterspace_capabilities_2025.pdf.
- ³¹ Andrew Imbrie, Owen Daniels, and Helen Toner, "Decoding Intentions: Artificial Intelligence and Costly Signals" Center for Security and Emerging Technology, October 2023, <https://cset.georgetown.edu/publication/decoding-intentions/>.
- ³² Chris Gordon, "China Practicing 'Dogfighting in Space,' US Space Force Says," *Air & Space Forces Magazine*, March 19, 2025, <https://www.airandspaceforces.com/china-practicing-dogfighting-space-force/>.
- ³³ Samson and Cesari, "Global Counterspace Capabilities."

³⁴ “In-Orbit Mission Success: Extending the Life of Intelsat 10-02,” Intelsat, June 21, 2021, <https://www.intelsat.com/resources/blog/in-orbit-mission-success-extending-the-life-of-intelsat-10-02/>.

³⁵ “Northrop Grumman Achieves First-Ever Undocking between Two Commercial Spacecraft in Geosynchronous Orbit,” Northrop Grumman, April 9, 2025, <https://news.northropgrumman.com/satellites/Northrop-Grumman-Achieves-First-Ever-Undocking-Between-Two-Commercial-Spacecraft-in-Geosynchronous-Orbit>; Brett Tingley, “A Chinese Satellite Just Grappled Another and Pulled It out of Orbit,” The War Zone, January 27, 2022, <https://www.twz.com/44054/a-chinese-satellite-just-grappled-another-and-pulled-it-out-of-orbit>.

³⁶ Victoria Samson, “Fact Sheet: Russian Military and Intelligence Rendezvous and Proximity Operations,” *Secure World Foundation*, December 2024, https://swfound.org/media/207995/fs24-03_russian-military-and-intelligence-rendezvous-and-proximity-operations.pdf.

³⁷ Loren Grush, “Russia Just Tested Satellite-Destroying Tech in Space, US Space Command Claims,” The Verge, July 23, 2020, <https://www.theverge.com/2020/7/23/21335506/russia-anti-satellite-weapon-test-kosmos-2543>.

³⁸ Matt Pyrak and Joseph Anderson, “Performance of Northrop Grumman’s Mission Extension Vehicle (MEV) RPO Imagers at GEO,” 2021, <https://amotech.com/TechnicalPapers/2021/Poster/Pyarak.pdf>; “RVS © 3000 Product Family Ready for the next Rendezvous the Advanced 3D Imaging LIDAR for Rendezvous and Docking. Acquisition, Tracking and Imaging of Both Cooperative and Non-Cooperative Targets. Space for Success.” Jena-Optronik, accessed May 1, 2025, https://www.jena-optronik.de/produkte/rendezvousensoren/rvs-3000.html?file=tl_files%2Fpdf%2FData+Sheet+RVS+3000+Product+Family.pdf; M Griebel, M Möller, S Chelkowski, and L Wagner, “GNC Sensors for in Orbit Servicing Missions,” Jena-Optronik, September 22, 2021, https://indico.esa.int/event/321/contributions/6361/attachments/4359/6581/210922_ESA-CSID_JOP-GNC-Sensors-for-in-Orbit-Servicing.pdf.

³⁹ List of tasks expands upon a list of functions from a 2023 NASA report on current technologies in space. Justin Goodwill, Christopher Wilson, and James Mackinnon, “Current AI Technology in Space,” NASA, July 2023, <https://ntrs.nasa.gov/api/citations/20240001139/downloads/Current%20Technology%20in%20Space%20v4%20Briefing.pdf>.

⁴⁰ Sumant Sharma, Connor Beierle, and Simone D’Amico, “Pose Estimation for Non-Cooperative Spacecraft Rendezvous Using Convolutional Neural Networks,” arXiv preprint arXiv:1809.07238 (2018), <https://arxiv.org/abs/1809.07238>.

⁴¹ Yucheng Chen, Yingli Tian, and Mingyi He, “Monocular Human Pose Estimation: A Survey of Deep Learning-Based Methods,” *Computer Vision and Image Understanding* Vol. 192, March 2020, <https://doi.org/10.1016/j.cviu.2019.102897>.

⁴² Tae Ha Park and Simone D'Amico, "Bridging Domain Gap for Flight-Ready Spaceborne Vision," arXiv preprint arXiv:2409.11661 (2024), <https://arxiv.org/abs/2409.11661v1>.

⁴³ Paul Day, Jeremiah Crane, Paul Cronk et al., "Experimental Results from On-Orbit Edge-Deployed AI Detection of Resident Space Objects Using Computer Vision," Advanced Maui Optical and Space Surveillance Technologies Conference, 2024, <https://amostech.com/TechnicalPapers/2024/Space-Based-Assets/Day.pdf>.

⁴⁴ Secretary of the Air Force Public Affairs, "Space Force Publishes 2025 Data, AI Strategic Action Plan," United States Space Force, March 19, 2025, <https://www.spaceforce.mil/News/Article-Display/Article/4124351/space-force-publishes-2025-data-ai-strategic-action-plan/>.

⁴⁵ Kelley Sayler, "Defense Primer: U.S. Policy on Lethal Autonomous Weapon Systems," Congressional Research Service, January 2, 2025, <https://www.congress.gov/crs-product/IF11150>; DOD Responsible AI Working Council, "U.S. Department of Defense Responsible Artificial Intelligence Strategy and Implementation Pathway," (Washington, DC: Department of Defense, June 2022), <https://media.defense.gov/2024/Oct/26/2003571790/-1/-1/0/2024-06-RAI-STRATEGY-IMPLEMENTATION-PATHWAY.PDF>.

⁴⁶ "Space Product Literature," BAE Systems, 2024, <https://www.baesystems.com/en-media/uploadFile/20210404061759/1434594567983.pdf>.

⁴⁷ AMD's Versal XQR chips are already radiation-hardened and suitable for space use off the shelf. Both NVIDIA chips are "harden-ready," meaning they will need to be equipped with additional shielding hardware prior to deployment aboard a spacecraft. "Aerospace & Defense Solutions | Space," Advanced Micro Devices, April 2, 2025, <https://www.amd.com/en/solutions/aerospace-and-defense/space.html>; Jason Black, "Boost Edge AI Performance with the New NVIDIA Jetson Orin NX 16GB," *NVIDIA Technical Blog*, January 24, 2023, <https://developer.nvidia.com/blog/boost-edge-ai-performance-with-the-new-nvidia-jetson-orin-nx-16gb/>; Leela Karumbunathan, "NVIDIA Jetson AGX Orin Series: A Giant Leap Forward for Robotics and Edge AI Application," NVIDIA, July 2022, <https://www.nvidia.com/content/dam/en-zz/Solutions/gtc/gtc21/jetson-orin/nvidia-jetson-agx-orin-technical-brief.pdf>.

⁴⁸ "Owning the Skies with Integrated Air Dominance," Lockheed Martin, January 22, 2025, <https://www.lockheedmartin.com/en-us/news/features/2025/owning-the-skies-with-integrated-air-dominance.html>.

⁴⁹ Andrew Lohn, "Hacking AI," Center for Security and Emerging Technology, December 2020, <https://cset.georgetown.edu/publication/hacking-ai/>.